

H. W. ALDEN.

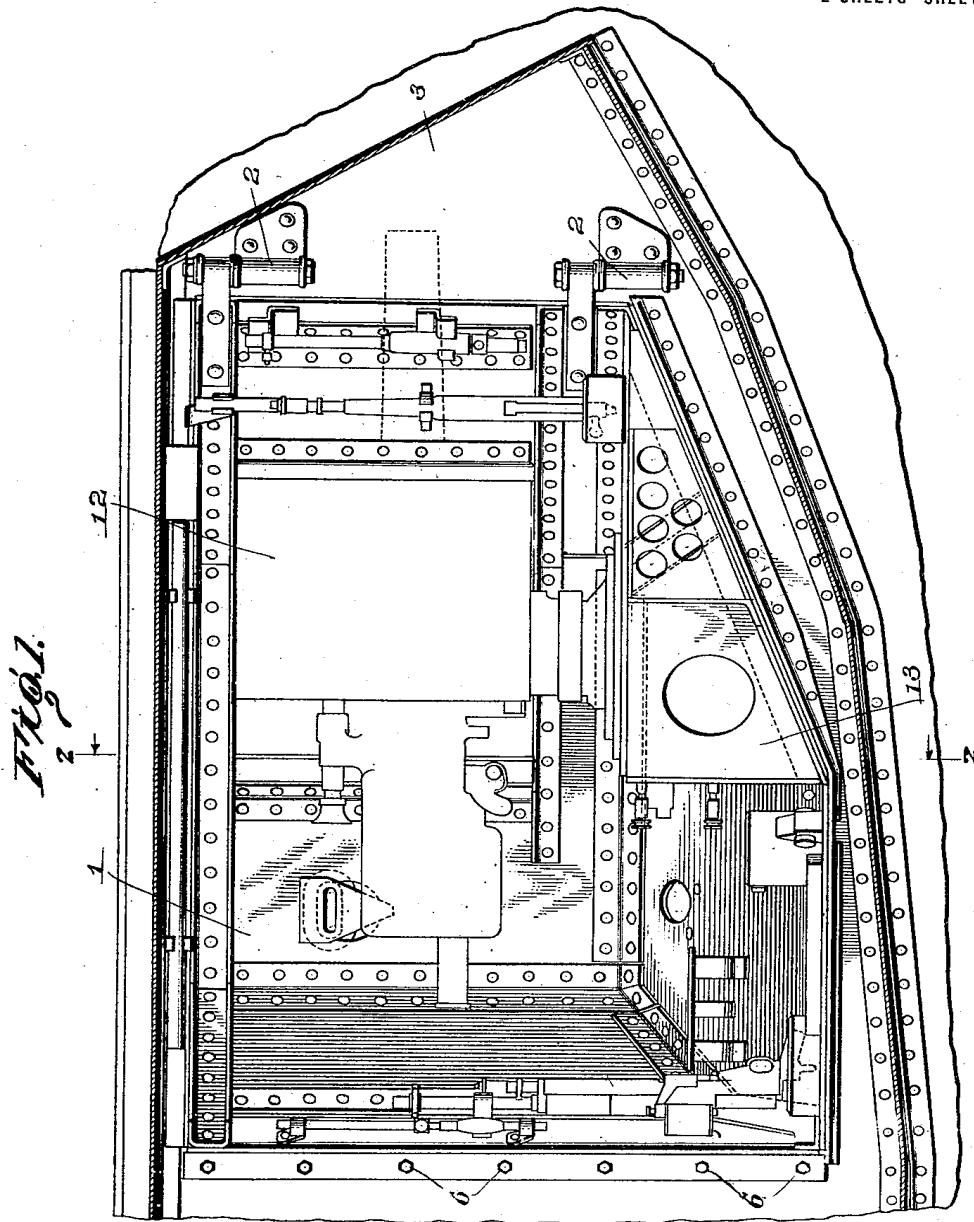
TANK.

APPLICATION FILED DEC. 14, 1918.

1,366,550.

Patented Jan. 25, 1921.

2 SHEETS—SHEET 1.



Inventor
Herbert W. Alden

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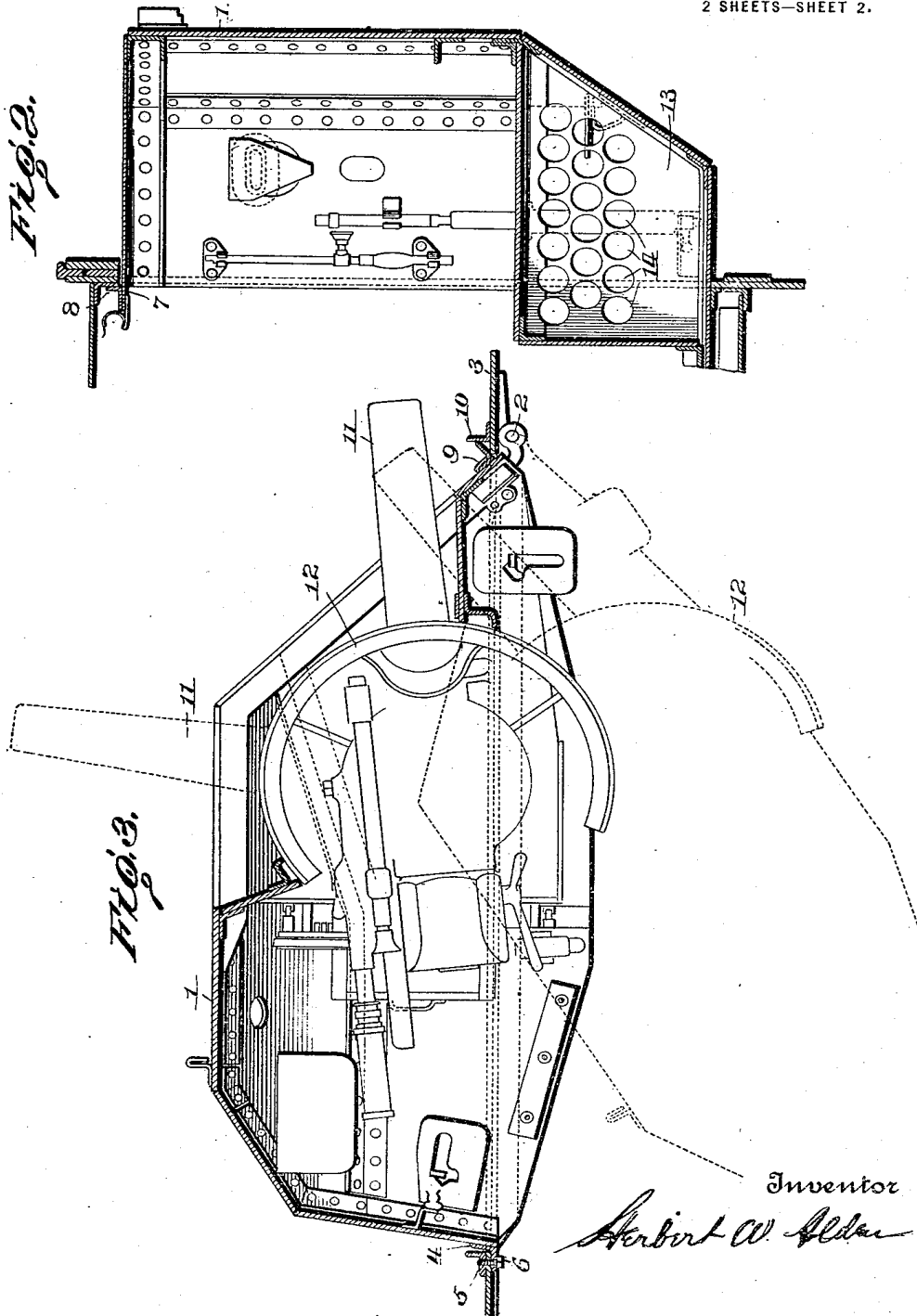
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Herbert W. Alden

UNITED STATES PATENT OFFICE.

HERBERT W. ALDEN, OF THE UNITED STATES ARMY, ASSIGNOR TO NEWTON D. BAKER, SECRETARY OF WAR OF THE UNITED STATES OF AMERICA, TRUSTEE.

TANK.

1,366,550.

Specification of Letters Patent.

Patented Jan. 25, 1921.

Application filed December 14, 1918. Serial No. 266,817.

(FILED UNDER THE ACT OF MARCH 3, 1883, 22 STAT. L., 625.)

To all whom it may concern:

Be it known that I, HERBERT W. ALDEN, lieutenant colonel, Ordnance Dept., United States Army, a citizen of the United States, stationed at Washington, District of Columbia, have invented an Improvement in Tanks, of which the following is a specification.

The invention described herein may be used by the Government, or any of its officers or employees in prosecution of work for the Government, or by any other person in the United States, without payment of any royalty thereon.

As military "tanks," so called, have heretofore been constructed, so far as I am aware, the sponson has been an immovable and irremovable structure, which projects from the side of the tank. In many instances, the sponson forms an obstruction to the passage of the tank, because (assuming a hypothetical case) it may be desired to drive the tank through a narrow ravine (artificial or natural) which is of insufficient width to permit the passage of the tank with such projecting sponson, but which is of a width which would permit the passage of a tank if such sponson were not present.

The principal object of my invention is, therefore, to provide a tank with a movable, projecting sponson, whereby a tank equipped with my invention may travel where tanks of proportionate size, but which are provided with immovable sponsons, cannot go, and thus accomplish results which it is impossible to attain with such other tanks.

With these objects in view, and others appearing as the specification proceeds and the nature of the invention is more fully disclosed, the invention resides, broadly and generally stated, in a tank provided with a movable sponson and, more specifically, in a tank provided with a sponson adapted to swing on a pivot from a position normally outside the tank to a position therewithin, to meet the exigencies of a particular situation.

The invention also resides in various structural refinements, tributary to the basic construction just referred to, which go to make up the ultimate perfection of the gun as an entirety.

The accompanying drawings disclose an exemplary concrete embodiment of the un-

derlying principles of my invention. Like reference characters identify corresponding parts throughout the several views, which latter may be briefly described as follows:

Figure 1 is a fragmentary longitudinal vertical sectional view of a tank equipped with a sponson constructed in accordance with my invention;

Fig. 2 is a cross sectional view on the line 2-2, Fig. 1; and

Fig. 3 is a horizontal sectional view through the sponson.

Referring, now, in detail to the drawings:

The sponson 1 is, in contra-distinction to the ordinary construction of sponsons, a movable body and, for this purpose, is appropriately supported upon hinges 2-2 carried by the tank frame 3, whereby the sponson may swing in a horizontal plane from the normal full-line position thereof to the dotted line position shown in Fig. 3, so as to occupy a position entirely within the tank.

Suitable means are provided for securing the sponson in its normal position exteriorly of the tank body or frame and while, conceivably, such means may take various forms, I have disclosed an exemplary means that may be used, and which I will now describe.

Carried by the marginal edge of that side of the sponson which is opposite the pivotal side thereof is an angle iron or bar 4, one of the flanges of such bar resting against the side of the sponson and the other flange constituting a lip which normally bears against the inner surface of the side wall 3 of the tank. The angle iron 4 thus also serves the purpose of limiting the outward movement of the sponson. Passing through the said lip and through said wall 3 and preferably through a strengthening member 5, which may be an angle iron, is a plurality of securing means, such as screws 6. It will be noted that, by the provisions of an angle iron 4, an effective closure is provided between the side of the sponson and the adjacent edge of the wall 3 of the tank, against entrance to the tank of dirt, moisture, etc.

The top of the sponson 1, in the normal position of the latter, is extended so as to project into the tank as shown in Fig. 2, thus forming, as it were, a lip 7; and carried on top of this lip is an angle iron or bar 8, one of the flanges thereof bearing against the

inner surface of the vertical wall of said tank, adjacent the opening through which the sponson projects. Said angle bar 8 may, obviously, of course, if desired, be secured to the front wall of the tank, instead of to the lip 7. This angle bar 8 affords a closure against entrance to the tank of dirt and moisture.

Adjacent the hinges 2, suitable means are provided for preventing the entrance to the tank of dirt and moisture, and in this instance, such means comprises an angle bar 9 carried by the sponson and contacting, in the normal position of the sponson, with a second angle bar 10 carried by the wall 3 of the tank.

The sponson 1 may carry a chamber 13 for the storage of shells. For the sake of lightness, the chamber is provided with numerous perforations 14.

While I have, in compliance with the Revised Statutes of the United States, described with great particularity one form of embodiment of my invention, it is to be understood that many changes will suggest themselves, particularly to those skilled in the art to which the invention pertains, and that all such changes as come within the scope of the appended claims constitute no departure from the spirit of the invention.

Having thus fully described my invention, what I claim as new and desire to secure by Letters Patent is:

1. In a military tank, a sponson normally disposed exteriorly of the tank and movable through a wall of the tank to a position within the tank, the sponson closing the wall when in normal position.

2. In a military tank, a pivotally mounted sponson normally disposed exteriorly of the tank and movable through an opening in the tank to a position therewithin, the sponson forming a closure for the opening when in normal position.

3. In a military tank, a sponson normally operating through and forming a closure for an opening in a wall of the tank, normally disposed exteriorly thereof and movable to a

position therewithin, and means for preventing movement of said sponson when positioned exteriorly of the tank.

4. In a military tank, a sponson operating through and forming a closure for an opening therein normally projecting therefrom and movable to a position therewithin, and means for securing said sponson in its normal position, including a member carried by said sponson and limiting outward movement thereof.

5. In a military tank, a sponson operating through and forming a closure for an opening in the tank, said sponson normally projecting therefrom and movable to a position therewithin, and means for securing said sponson in its normal position, including an angle iron carried by said sponson and limiting outward movement thereof.

6. In a military tank, a sponson movably operating through and forming a closure for an opening in the tank, said sponson normally projecting therefrom and movable to a position therewithin, means for securing said sponson in its normal position, including an angle iron carried by said sponson and limiting means passing through said angle iron and the wall of said tank.

7. A disappearing sponson for tanks pivoted at its forward end and operating through an opening in a wall of the tank, and having a protecting shield forming a complete closure to the opening when the sponson is in normal position.

8. A disappearing sponson for tanks or the like adapted to operate through an opening in the side of said tank and supporting a piece of ordnance and forming a closure to the tank when in its outermost position.

9. A disappearing combination sponson and gun mount adapted to operate through an opening in an armored wall, pivoted at its forward edge and forming a closure in the wall when swung to its outermost position.

HERBERT W. ALDEN.